

Reliable Agent-Driven Development

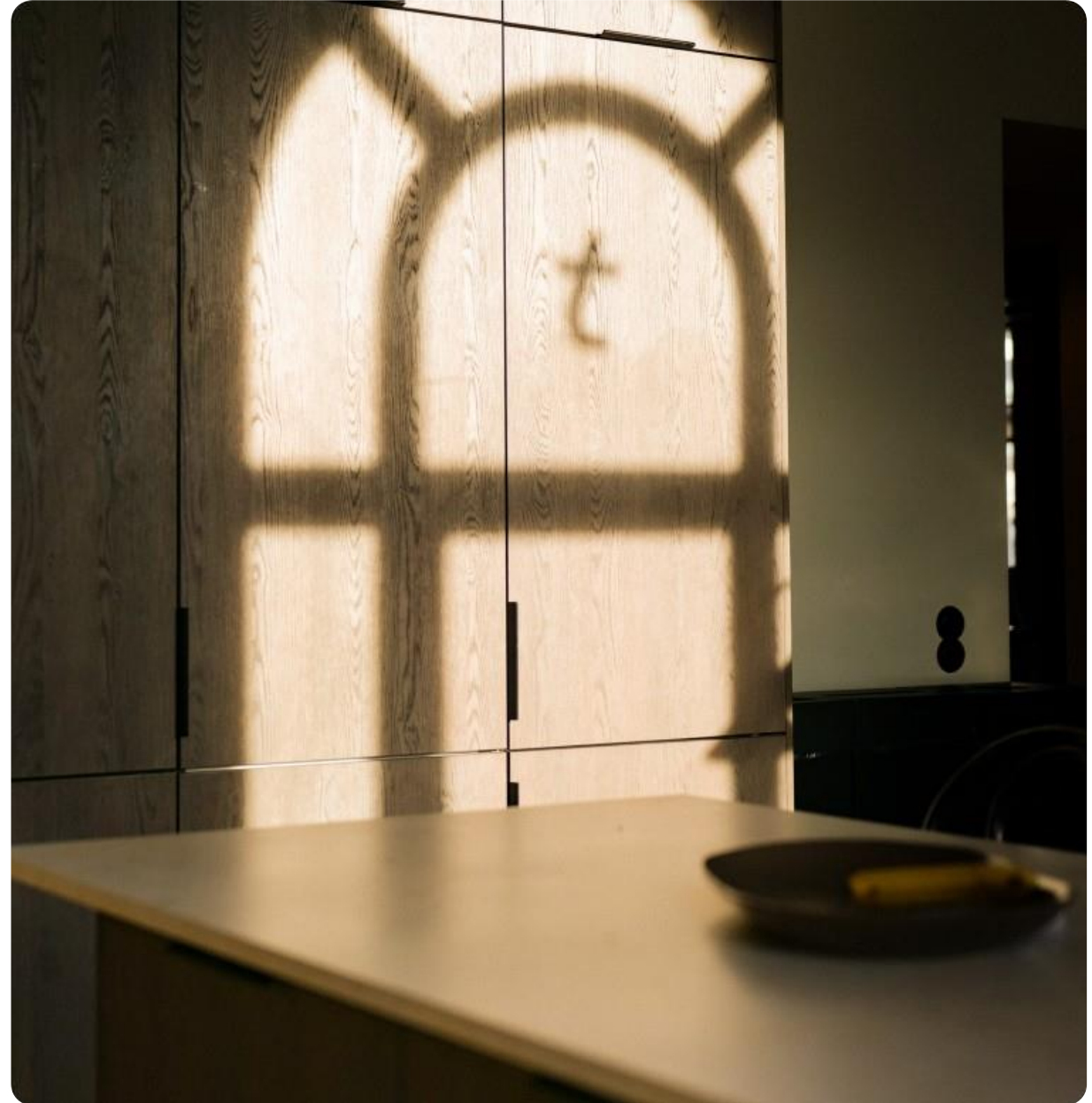
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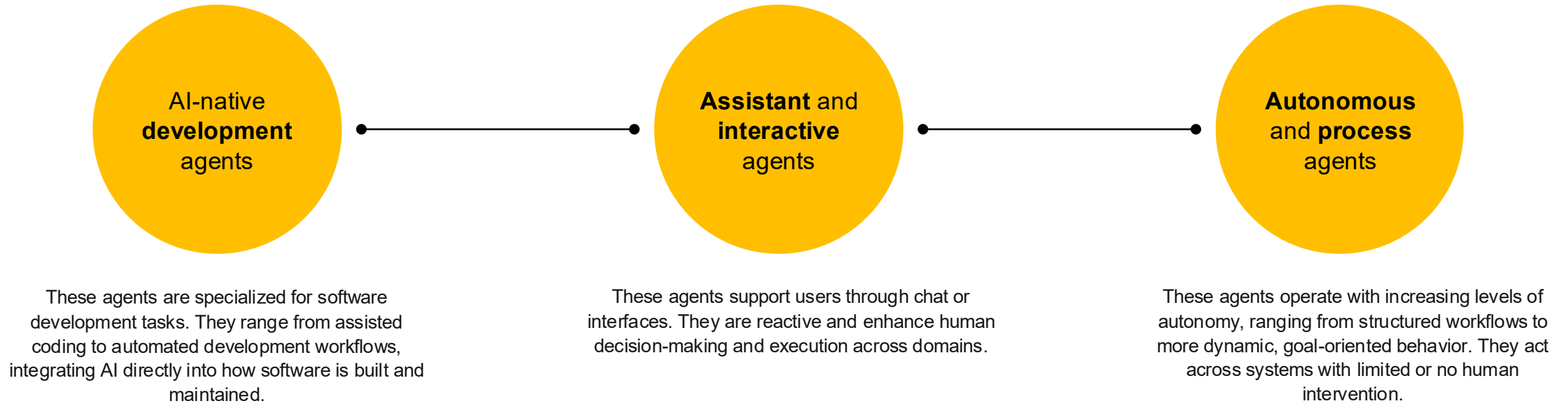
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Introduction



Agentic **design** and **control** in practice



The risks of using AI in development

So much for 'trust but verify': Nearly half of software developers don't check AI-generated code – and 38% say it's because it takes longer than reviewing code produced by colleagues

A concerning number of developers are failing to check AI-generated code, exposing enterprises to huge security threats

The Register

AI coding tools make developers slower but they think they're faster, study finds

Predicted a 24% boost, but clocked a 19% drag

 [Thomas Claburn](#)

Fri 11 Jul 2025 // 22:41 UTC

Using AI might actually slow down experienced devs

News By Craig Hale published July 11, 2025

Experienced developers aren't getting any benefit from AI, report claims

Developers Lean on AI More, But Report Growing Doubts About Accuracy, Stack Overflow Survey Says

The Risks of Using AI in Software Development

Why AI Coding Tools Make Experienced Developers 19% Slower and How to Fix It

Oct 3, 2025  Molisha Shah

Different levels of agentic engineering



Prototype, proof-of-concepts or explorative solutions with no focus on security

Build using vibe-coding tools like Lovable, GitHub Spark, Figma Make, etc.

Not recommended for production but great for testing ideas, working on different workflow, A/B testing. Can be done by most people in the organization without any technical knowledge including knowledge about enterprise security, governance, etc.



Pilots, MVPs, or small internal tools with minimum focus on scalability and governance

Build using pro-coding tools combined with agentic tools like GitHub Copilot, Claude Code, etc. and using structured knowledge, workflows, and reusable skills

Ready for production but with limited focus on security, governance, and scalability. Build using structured knowledge, reusable components and skills. Development is handled by agents but validated by developers. Optimal for small to medium greenfield and brownfield projects.



Structured, predictable, scalable, and governed enterprise solutions

Build using pro agentic tools combined using structured knowledge, workflows, reusable skills, and enterprise governance patterns

Ready for production with extensive focus on security, governance, and scalability. Build using structured knowledge, workflows, reusable components, and skills. Development is handled by agents but validated by developers. Optimal for medium to large greenfield and brownfield projects with mixed tech stack including modernization.

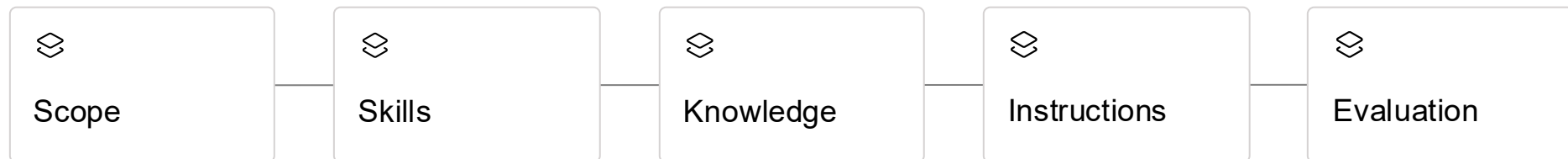
Agentic **design** and **control** in practice



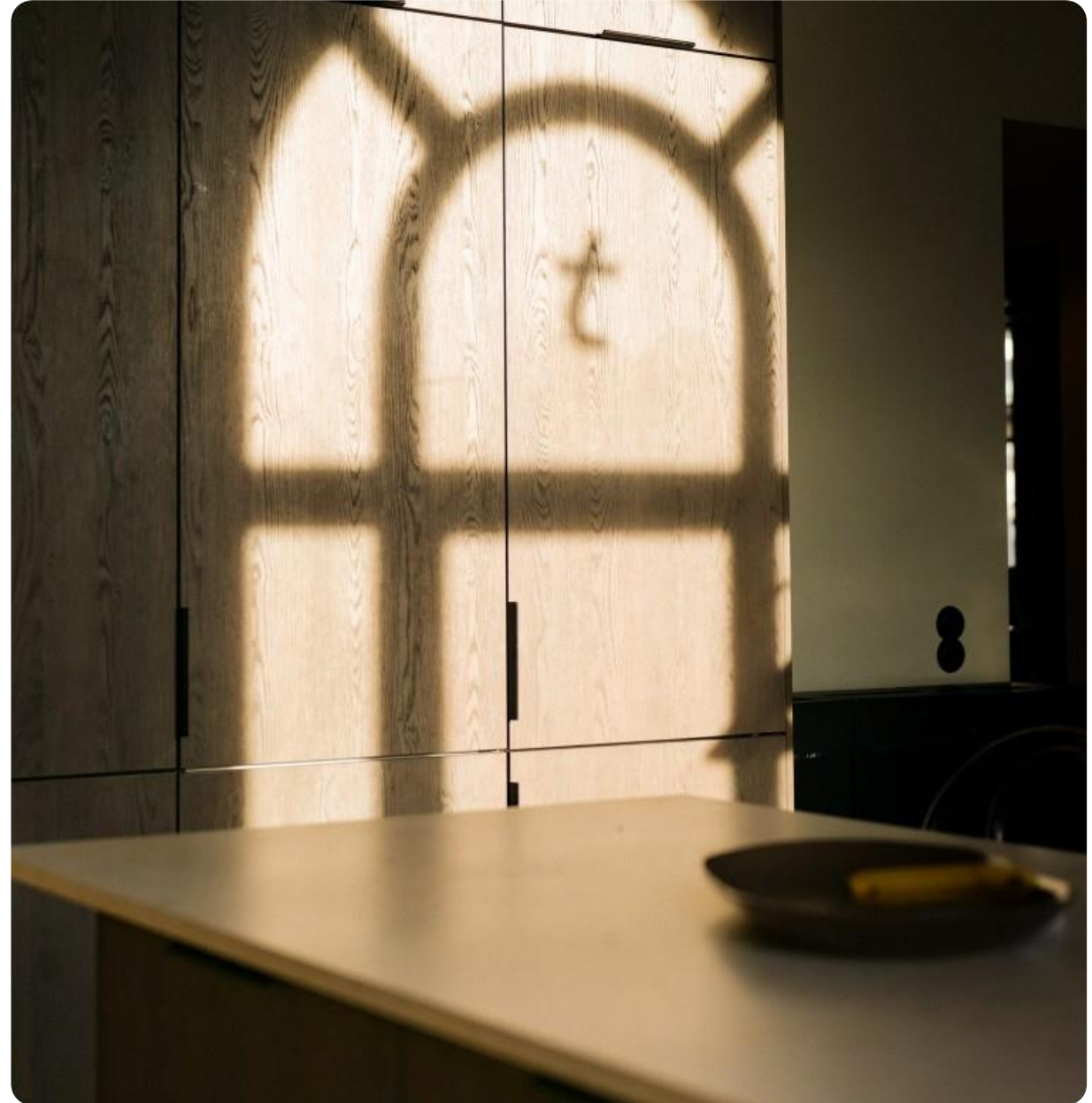
These agents are specialized for software development tasks. They range from assisted coding to automated development workflows, integrating AI directly into how software is built and maintained.

These agents support users through chat or interfaces. They are reactive and enhance human decision-making and execution across domains.

These agents operate with increasing levels of autonomy, ranging from structured workflows to more dynamic, goal-oriented behavior. They act across systems with limited or no human intervention.

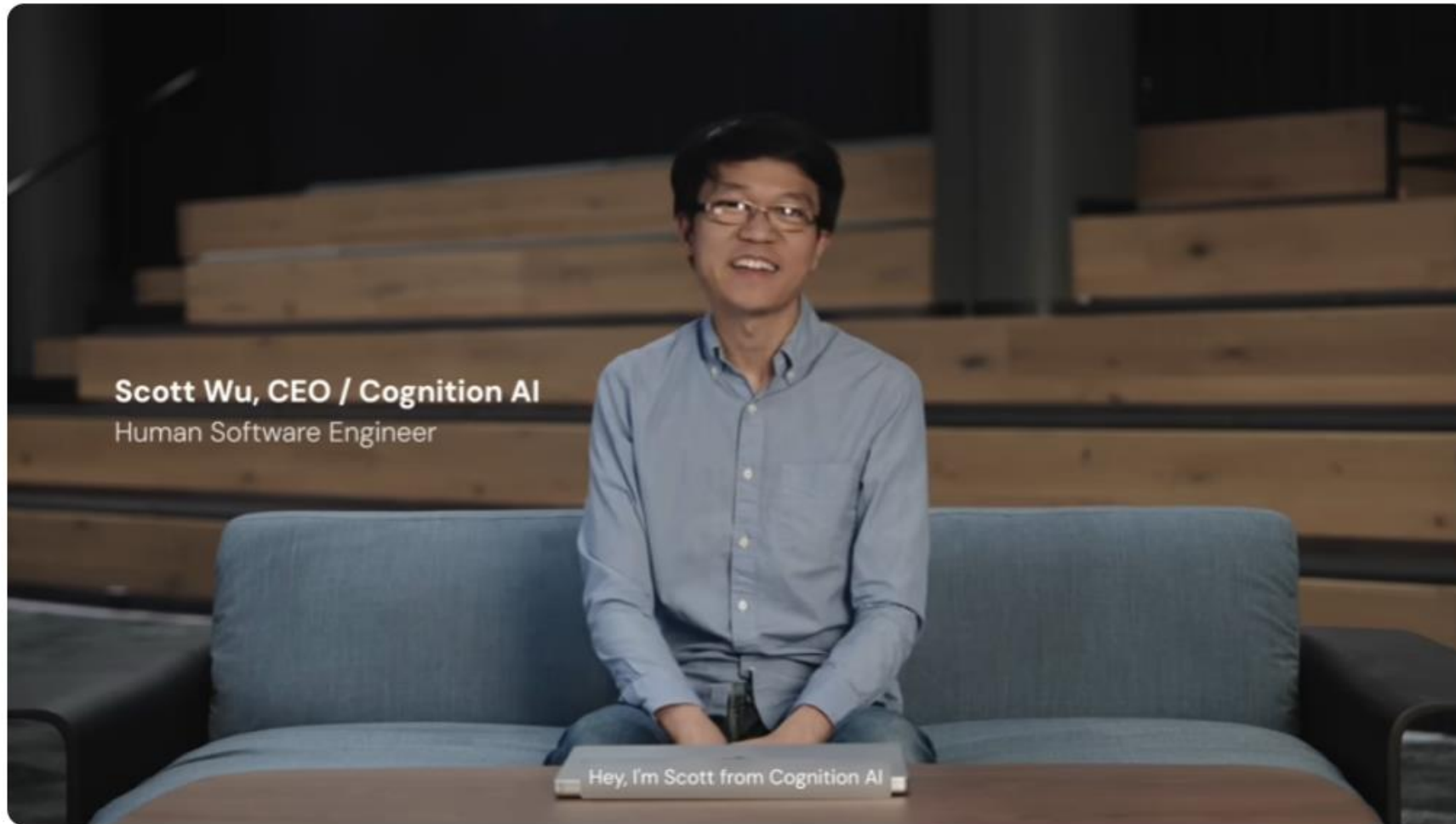


Agentic Development



A man with dark hair and glasses, wearing a light blue button-down shirt, is sitting on a blue sofa. He is smiling and looking towards the camera. In front of him is a wooden table with a silver laptop on it. The background consists of wooden slats arranged in a stepped pattern, resembling bleachers or a modern architectural wall.

Scott Wu, CEO / Cognition AI
Human Software Engineer



Scott Wu, CEO / Cognition AI
Human Software Engineer

Introducing Devin, the first AI software engineer



Cognition
28.4K subscribers

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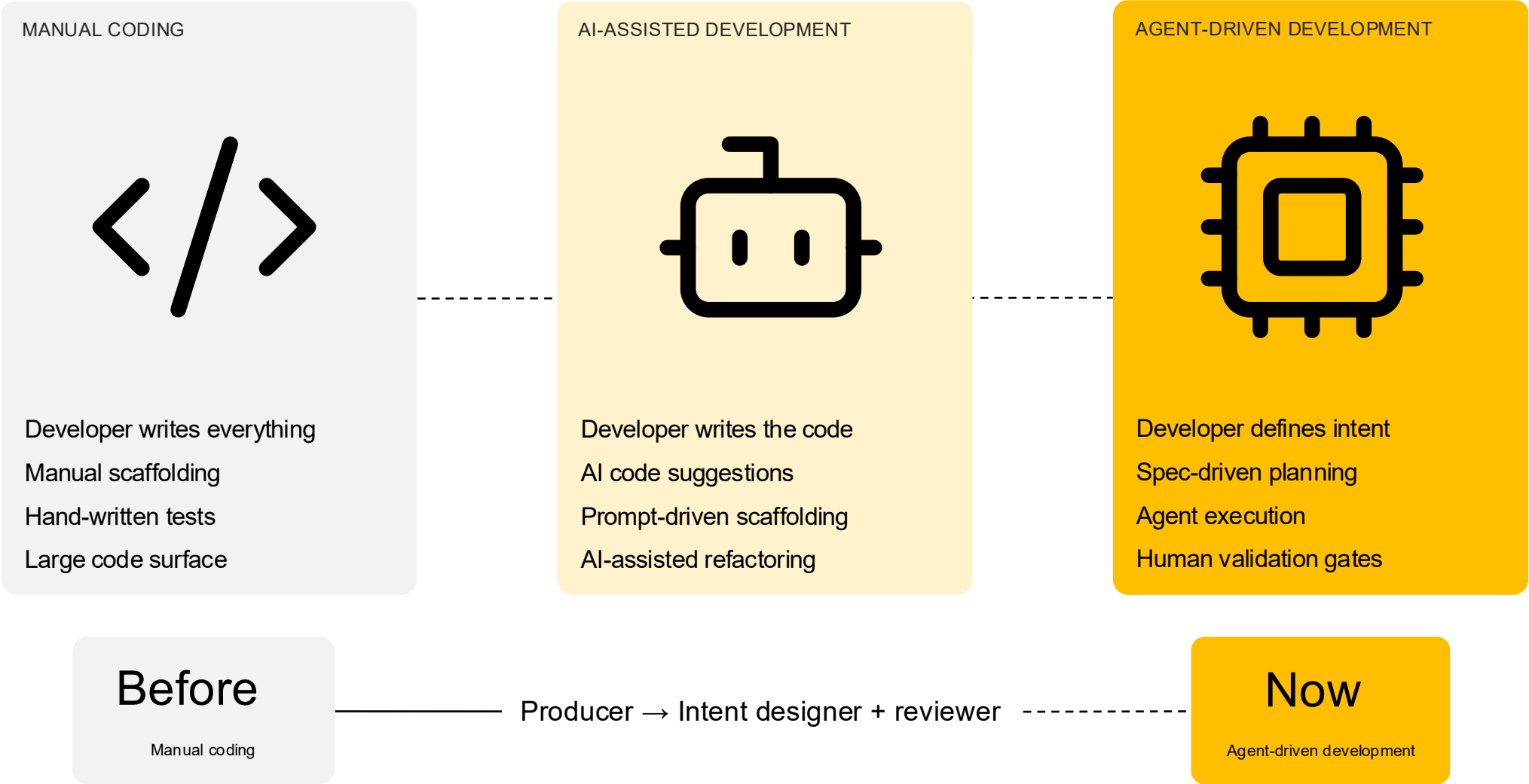


1.2M views 1 year ago

Meet Devin, the world's first fully autonomous AI software engineer.

Devin is a tireless, skilled teammate, equally ready to build alongside you or independently complete tasks for you to review. ...more

How development changes

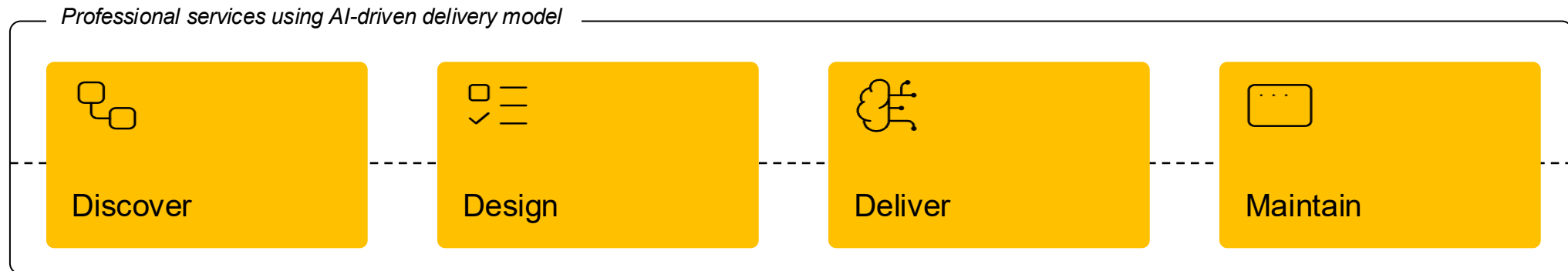


The way we deliver and
operate is in change

Delivery of enterprise software solutions using an agent-driven approach

Roles

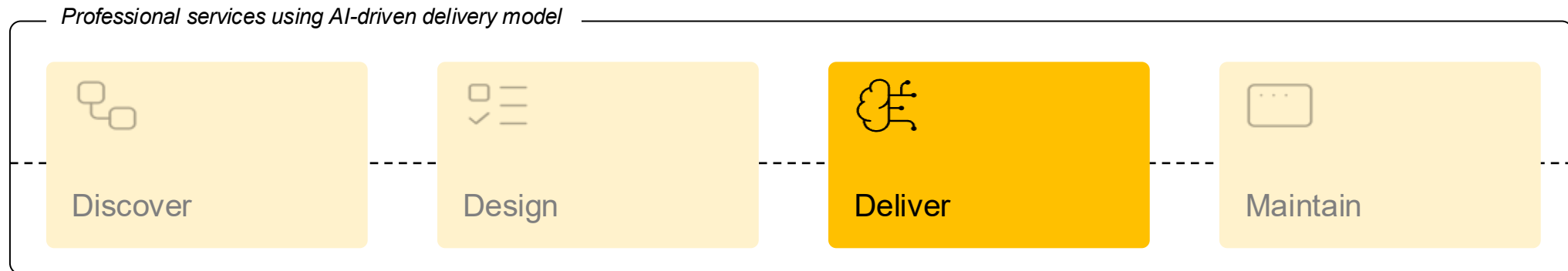
Innovation lead, Discovery facilitator, UX specialist, AI engineer, QA approver, Project orchestrator



Delivery of enterprise software solutions using an agent-driven approach

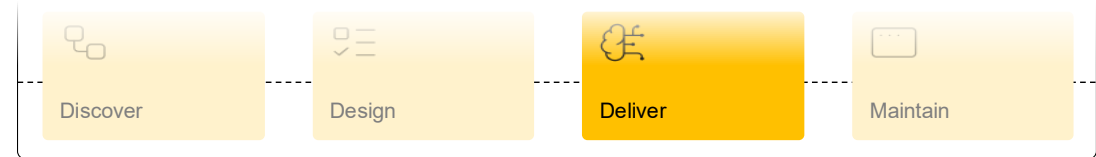
Roles

Innovation lead, Discovery facilitator, UX specialist, AI engineer, QA approver, Project orchestrator

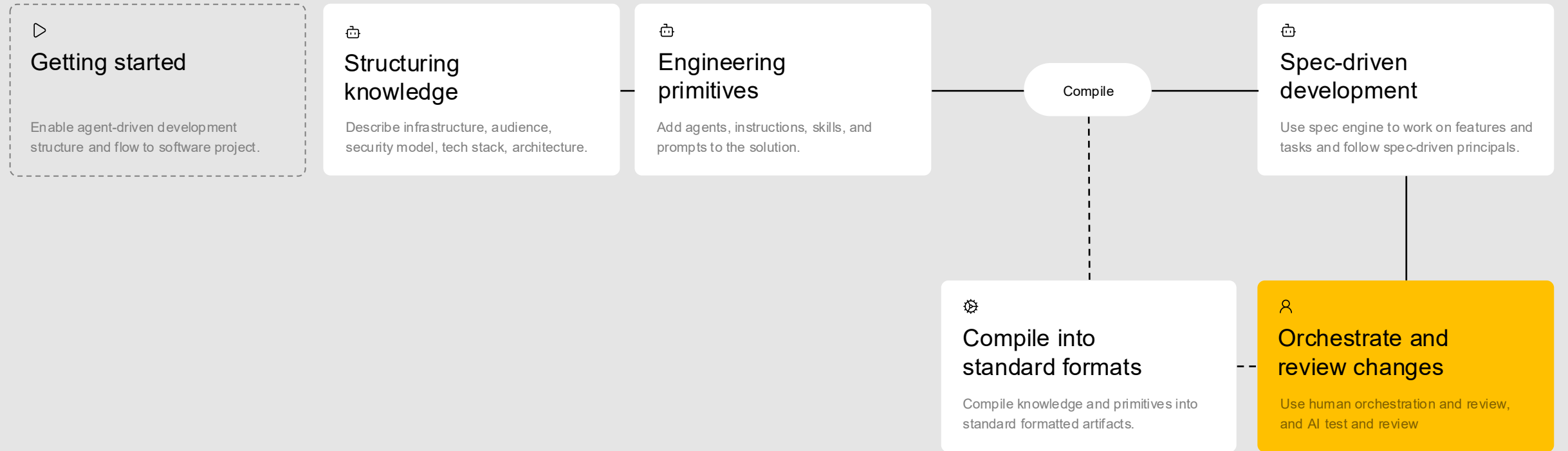


Deliver using agent-driven development

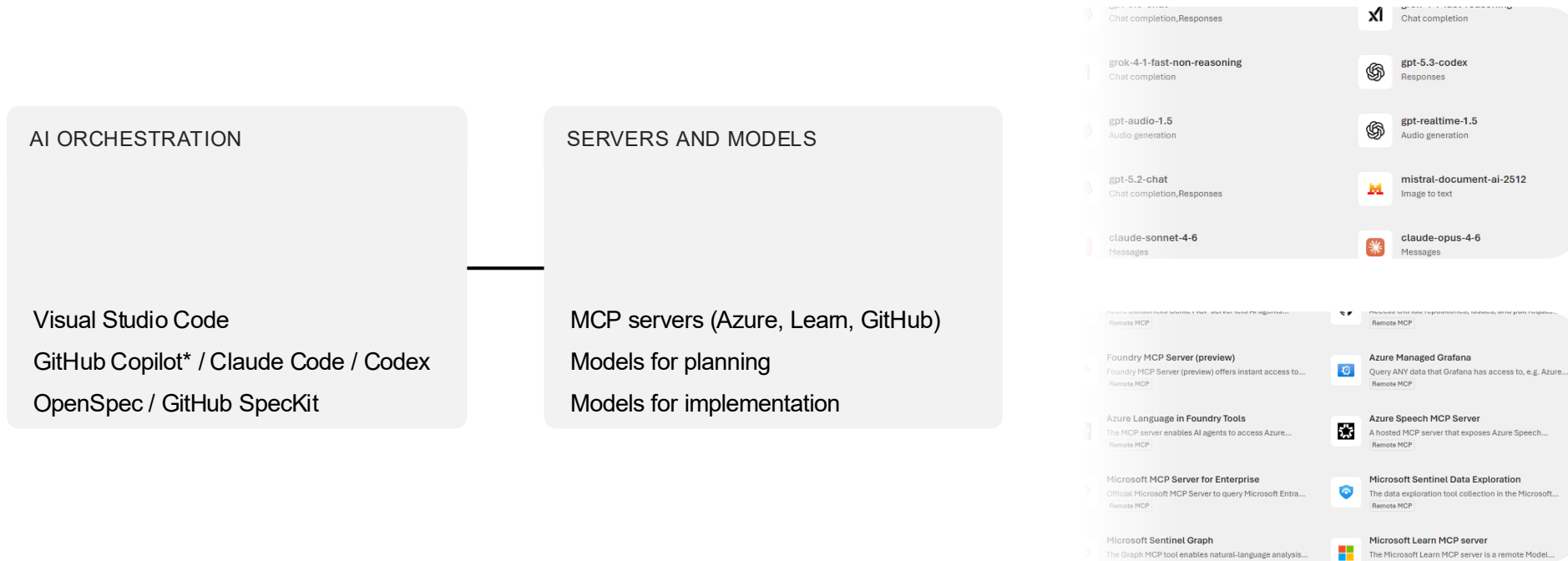
Structured, AI-orchestrated processes where the developer defines intent, specifications guide planning, agents execute tasks, and human validation gates ensure control and quality across the end-to-end flow. Designed to be predictable, reusable, team-scalable, and governed.



DEVELOPMENT FLOW



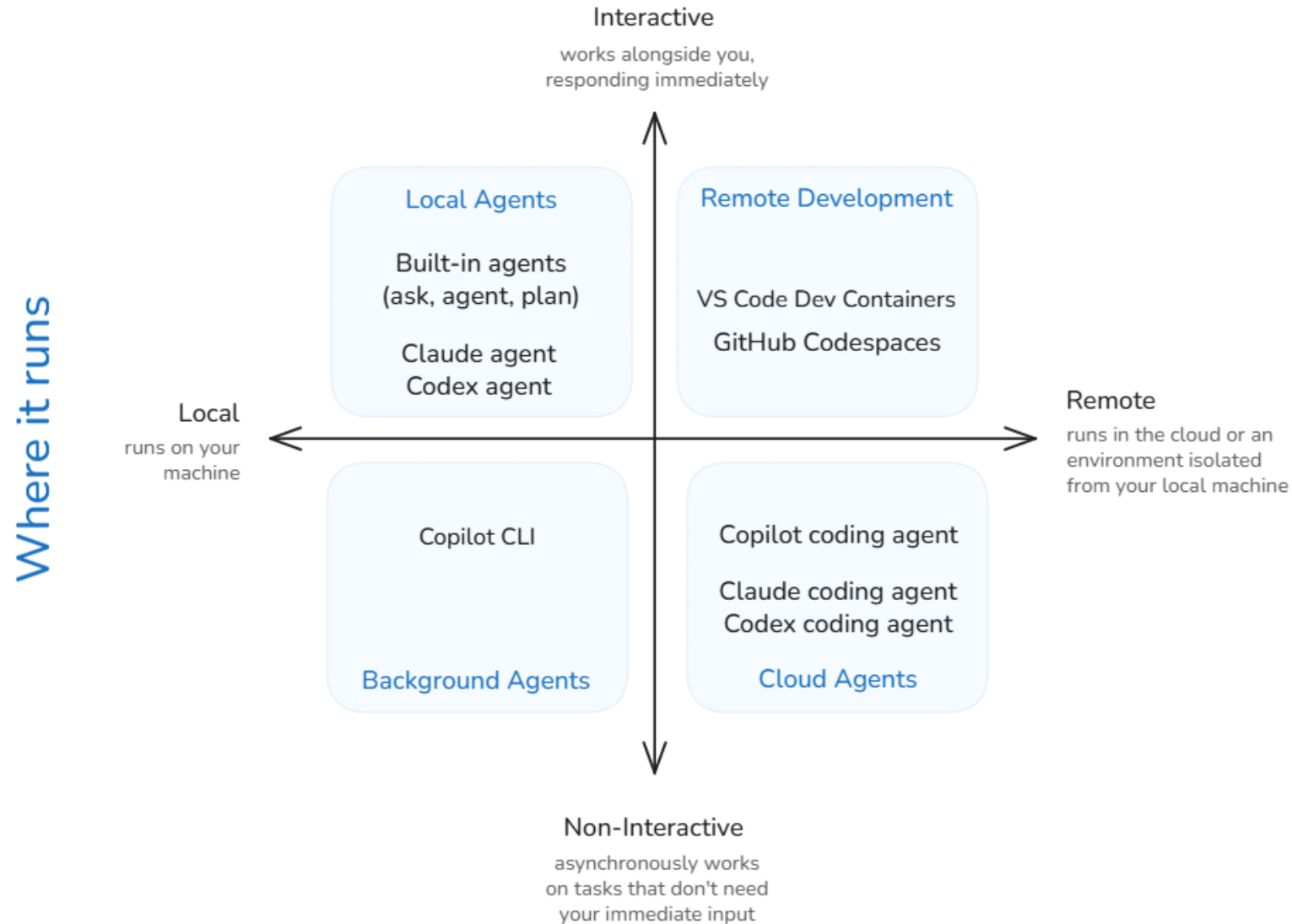
Stay on top of your AI tools and resources



* Business (300 PRPM) or Enterprise (1500 PRPM)

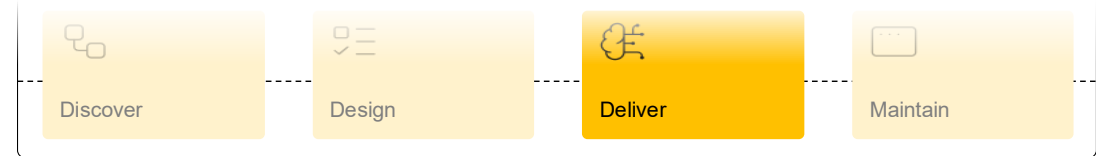
Agent sessions

How you collaborate

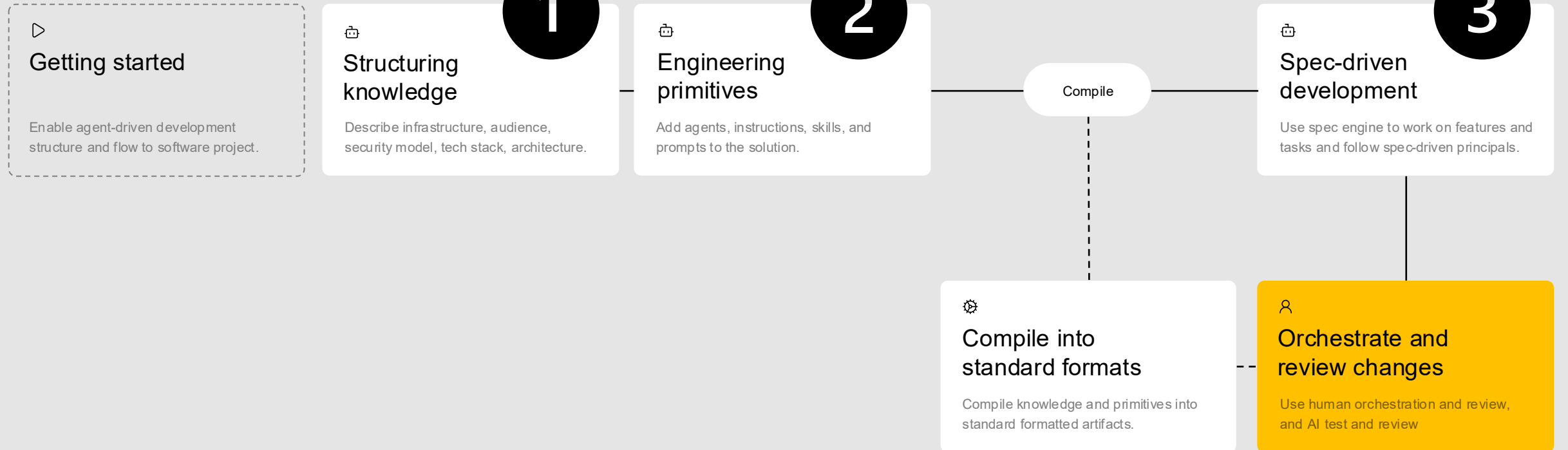


Deliver using agent-driven development

Structured, AI-orchestrated processes where the developer defines intent, specifications guide planning, agents execute tasks, and human validation gates ensure control and quality across the end-to-end flow. Designed to be predictable, reusable, team-scalable, and governed.



DEVELOPMENT FLOW



Demo: Structuring knowledge

KNOWLEDGE

Architecture

Audience

Roles

UX/UI

Security

Tech stack

Describe infrastructure, audience, security model, tech stack, architecture.

.github/copilot-instructions.md

openspec/config.yaml

Tech Stack

Backend

- .NET 10 — runtime and framework target
- ASP.NET Core 10 — web host
- FastEndpoints — HTTP endpoint handler framework (replaces Controllers)
- Entity Framework Core + Npgsql — ORM + PostgreSQL driver
- OpenAI SDK v2.x (OpenAI NuGet) — OpenAI chat interactions
- Azure AI Projects SDK (Azure.AI.Projects) — Microsoft Foundry agent/chat/flow interactions
- Model Context Protocol (ModelContextProtocol NuGet) — MCP server tools
- .NET Aspire 13.1 — orchestration and service composition
- xUnit — unit and integration tests
- System.Text.Json — serialization (Newtonsoft.Json is banned)

Frontend

- React 19 — UI framework
- TypeScript 5 — language
- Vite 7 — build tooling and dev server
- TanStack Query 5 — server state management
- React Router 7 — client-side routing
- Shadcn/UI + Tailwind CSS — component library and styling
- React Hook Form + Zod — form handling and runtime validation
- VS Code Codicons — icon font for activity bar glyphs

Infrastructure / Platform

- Azure Database for PostgreSQL Flexible Server — primary database
- Azure Cache for Redis — caching layer
- Azure Blob Storage — file/content storage
- Azure Key Vault — secrets management
- Microsoft Entra ID — authentication
- Azure Container Apps — container hosting
- Azure Monitor + Application Insights — telemetry and monitoring

Structuring knowledge

Describe infrastructure, audience, security model, tech stack, architecture.

Er

pr

Add

pror

Demo: Engineering primitives

PRIMITIVES

Agents

Instructions

Skills

Prompts

Plug-ins

Hooks

Add agents, instructions, skills, and prompts to the solution.



`.github/agents/*`
`.github/instructions/*`
`.github/skills/*`

Agents

Name	Filename	Description
Architect [exists]	<code>architect.agent.md</code>	Defines end-to-end architecture and contracts for the platform: API design, layer boundaries, cross-cutting concerns, security approach, and deployment topology.
Backend Developer [exists]	<code>developer-backend-dotnet.agent.md</code>	Implements the .NET 10 backend using Clean Architecture, Aspire, FastEndpoints, EF Core, and PostgreSQL. Handles REST API endpoints, auth validation, and unit/HTTP tests.
Frontend React Developer [exists]	<code>developer-frontend-react.agent.md</code>	Implements the React 19 + TypeScript 5 + Vite 7 frontend. Delivers UI flows, TanStack Query integration, React Router navigation, Shadcn/UI components, and Tailwind CSS styling.
Reviewer [exists]	<code>reviewer.agent.md</code>	Reviews specs, API contracts, implementation code, and documentation for correctness, architecture alignment, and governance conformance. Produces structured, actionable feedback.
Security Review [exists]	<code>security-review.agent.md</code>	On-demand security reviewer. Scans source code, configuration, infrastructure, and CI/CD workflows for OWASP Top 10 weaknesses. Provides prioritized remediation guidance.



Engineering primitives

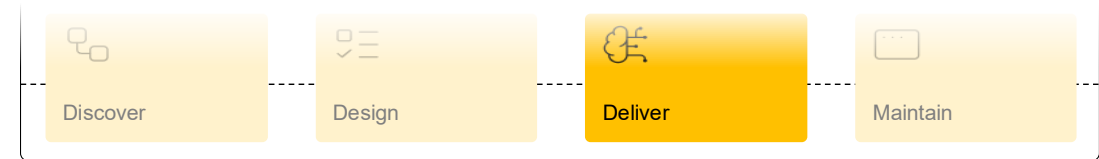
Add agents, instructions, skills, and prompts to the solution.

agent = role
instructions = guidance and norms
prompt = current task
skill = reusable capability
plugin = installable package of capabilities
hook = event-driven automation around execution

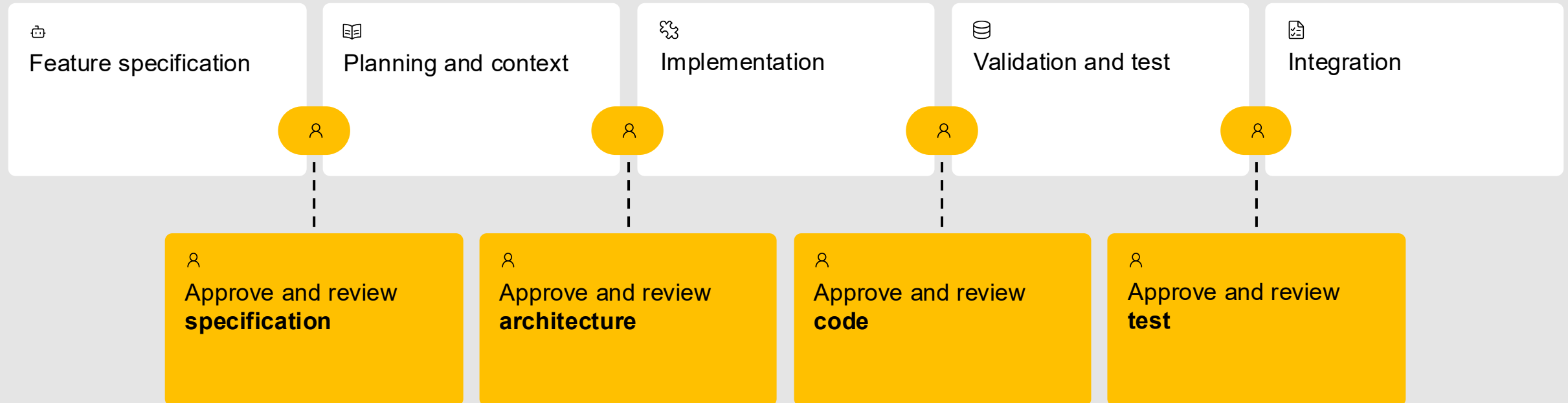
Spec-driven development

Follow spec-driven principals: Change => Implement => Archive.

Use human orchestration and review, and AI test and review



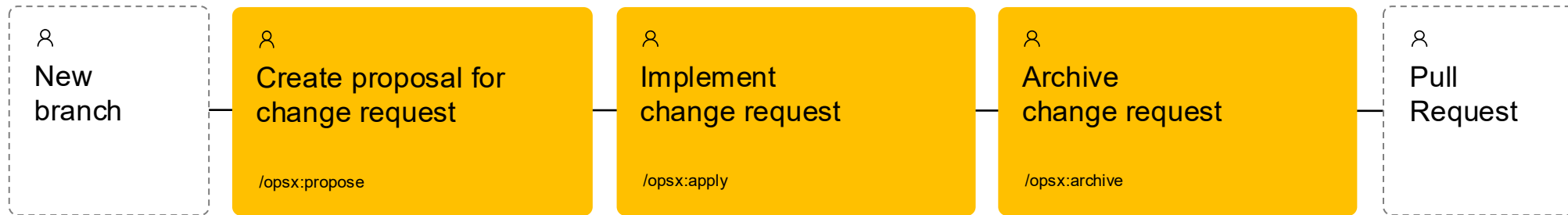
SPEC-DRIVEN DEVELOPEMENT



Demo: Spec-driven development



Demo: Spec-driven development



Agentic building blocks

Primitives are the core configurable elements that developers iteratively refine to ensure reliable outcomes.

AGENTIC BUILDING BLOCKS



Engineering primitives

Add agents, instructions, skills, and prompts to the solution.



Agents

Reusable agents that orchestrate the role, permissions and operating boundaries.



Instructions

Scoped rules and the standards that the agent must follow and in which context.



Skills

Capabilities the agent can use within specific areas and context windows.



Hooks

Capabilities the agent can use within specific areas and context windows.



Orchestration

Use spec engine to work on features and tasks and follow spec-driven principals.



Specifications

Source of truth covering intent, constraints and acceptance criteria.



Memory

Decisions, notes and state maintained and carried forward.



MCP tools

Use spec engine to work on features and tasks and follow spec-driven principals.



Plugins

Decisions, notes and state maintained and carried forward.

Predictable, reusable, and scale across teams

Development time
decrease with

+50%

**01
10**

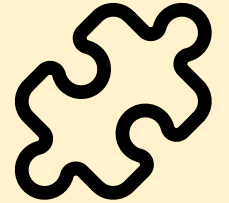
Quality and
security up with

+20%



Improve
standardization

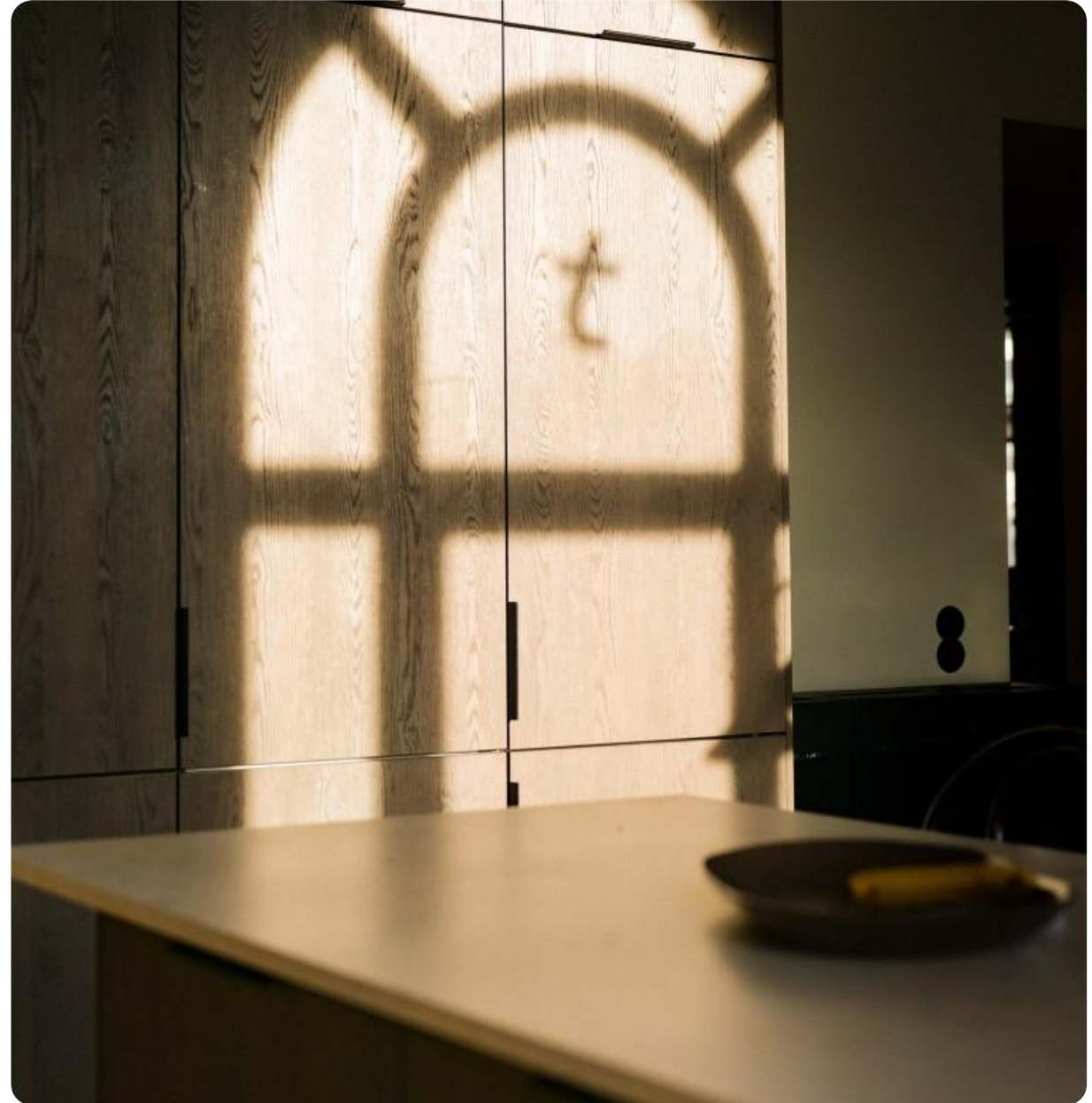
>50%



Happy developers

+20%

Questions



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